Design and fuction

Contents

General structure of the machine

Base frame	The machine is mounted on a base frame made of rustproof stainless steel with vibration-damping, variable-height feet.
Paneling	The machine is equipped with protective doors with protective switches and by a stainless steel housing with locks in the areas for the drive and installation. For further information on the safety devices, please refer to chapter "Safety".
Production-, drive- and installation areas	The production area is separated from the drive and installation areas by a plate. This means that the electrical and pneumatic equipment on the machine are housed in the rear or lower part of the machine depending on the machine type.

Limits of the machine

Attention!

If hazardous, easily combustible/flammable materials are processed, the system owner is to ensure:

- safe processing,
- extraction device in the working area.

Fundamentals of the machine design

Principle

The machine/system is constructed according to the state of technology and the recognized safety rules. Nevertheless, when used, dangers may arise for the life and limb of the user or third parties or interference may affect the machine and other property.

Using the machine

The machine must only be operated in perfect technical working order as well as intended, with attention given to safety and potential dangers under observation of the operating instructions! In particular, immediately rectify (or have rectified) malfunctions which could affect safety!

Intended use

Application of the machine

The machine is suitable for mounting tips and caps with one another and to shrink-wrap them together with infed catheters in film.

The machine may only be operated as intended with the products and/or formats listed in the corresponding business order. Only personnel who are qualified to perform their intended task may work at the machine (see operating personnel / user groups).

Any other use is considered a non-intended use. The manufacturer is not liable for any damages resulting from non-intended use; the system owner bears responsibility in this case.

Intended use also includes the adherence to the operating, maintenance and repair conditions specified by the manufacturer.

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Residual risk

In spite of all precautions, the following residual risk remains:

Attention!

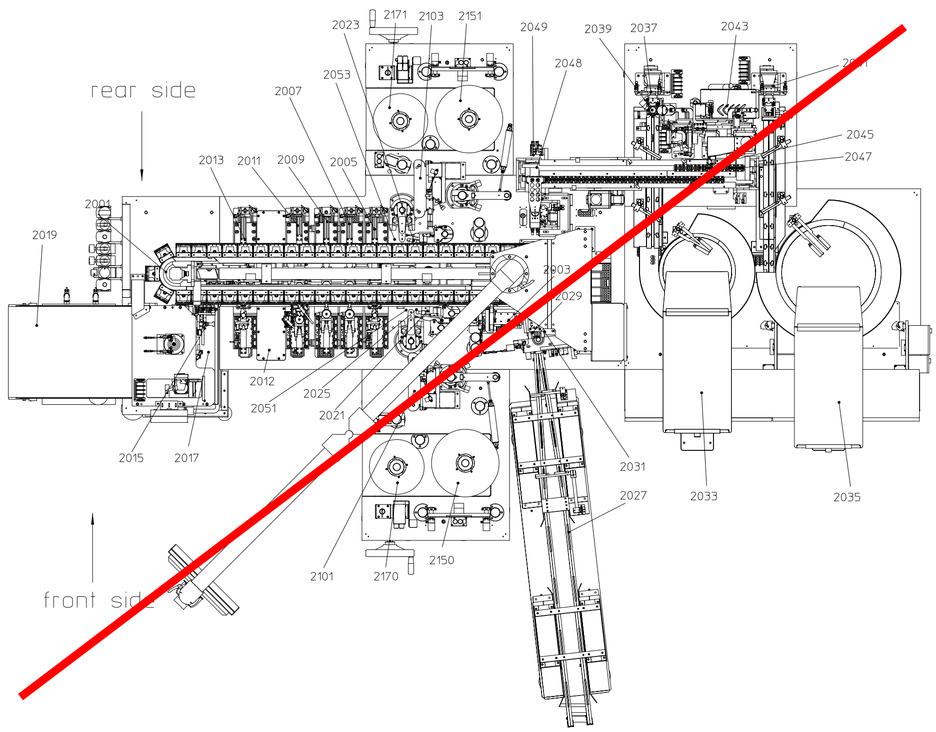
Risk of injury.
Protective doors are bridged when in Setup operation. As a result, it is possible to perform machine movements while the protective doors are open.
Only authorized and trained persons who are familiar with the machine operation and are aware of the possible dangers may bridge protective doors.

- In Setup operation, the machine must be operated by only one person.
- Secure the machine against accidental operation by another person, e.g., affix warning sign.

Warning!

Risk of crushing.
Tools can crush hands and fingers.
Do not reach into the station tools.

Machine overview



18365_Kap5_Stationen.ai

The machine consists of the following stations:

Station 2001 - Oval Turret

Station 2003 - Insert Catheter in Tips

Station 2005 - Tack Sealing Funnel & Tip

Station 2009 - Free Station

Station 2011 - Long Seal

Station 2012 - Foil Separation

Station 2013 - Funnel & Tip Sealing

Station 2015 - Pick & Place Catheters

Station 2017 - Reject Station

Station 2019 - Discharge Conveyor

Station 2021 - Splice Detection Front Film

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Station 2023 - Splice Detection Rear Film

Station 2025 - Free Station

Station 2027 - Punched Catheter Feeding

Station 2029 - Punched Catheter Transfer

Station 2031 - CH Verification

Station 2033 - Tip Feeding

Station 2035 - Cap Feeding

Station 2037 - Insert Tip

Station 2039 - Vision System Tip

Station 2041 - Insert Cap

Station 2043 - Cap/Tip Assembly (incl. Lubragel Container)

Station 2045 - Cap/Tip Transfer

Station 2047 - Load pucks/buffer

Station 2048 - Unload pucks

Station 2049 - Insert Cap/Tip

Station 2051 - Position Verification Front Film

Station 2053 - Position Verification Rear Film

Station 2101 - Edge Guiding Front Film

Station 2103 - Edge Guiding Rear Film

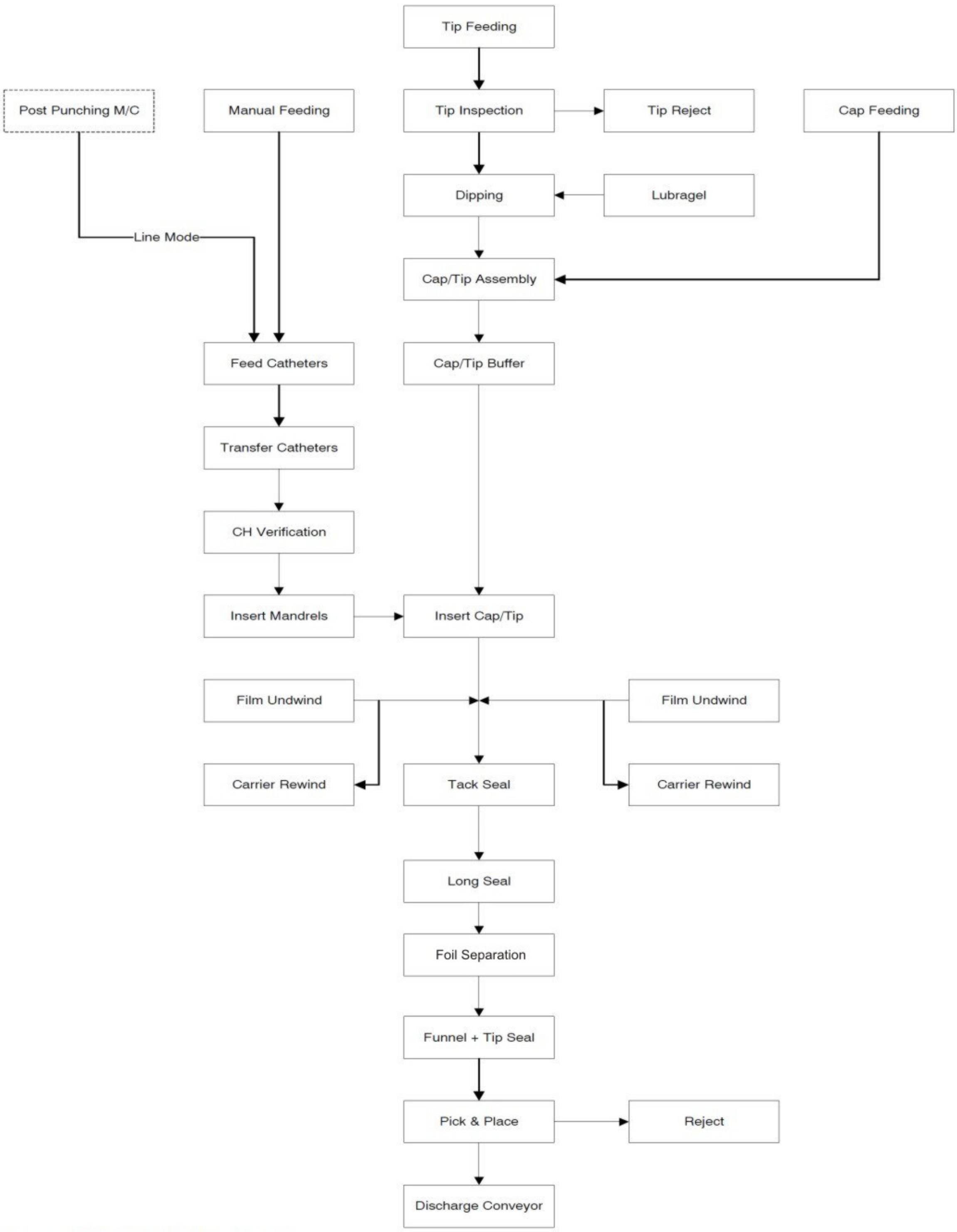
Station 2150 - Unwind Front Film

Station 2151 - Unwind Rear Film

Station 2170 - Rewind Front Film Carrier

Station 2171 - Rewind Rear Film Carrier

Function description in regular operation



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Function description for regular operation

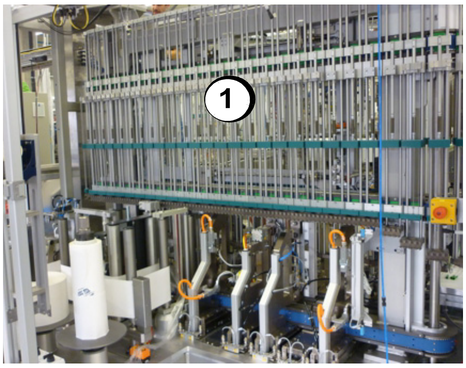
Station 2001 - Oval Turret

The oval turntable is vertically oriented and mounted on the machine plate. It is driven by three, servo-controlled toothed belts:

- the upper belt holds the clamping plate arbor
- the middle belt holds the guide pusher for the funnel containers.
- the lower belt holds the workpiece carriers of the cap/tip pucks.

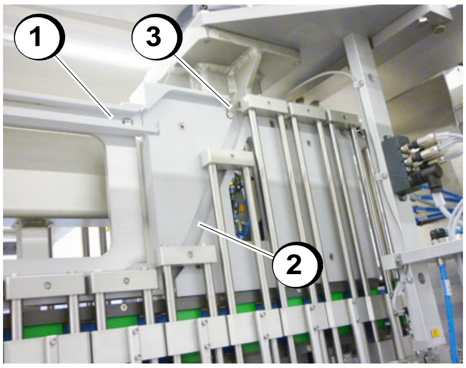
One workpiece carriers is assembled with four holding fixtures.

A format change between 20 cm and 40 cm is performed by pneumatically switching between two cam tracks.



4478002B001

- Turntable (1)



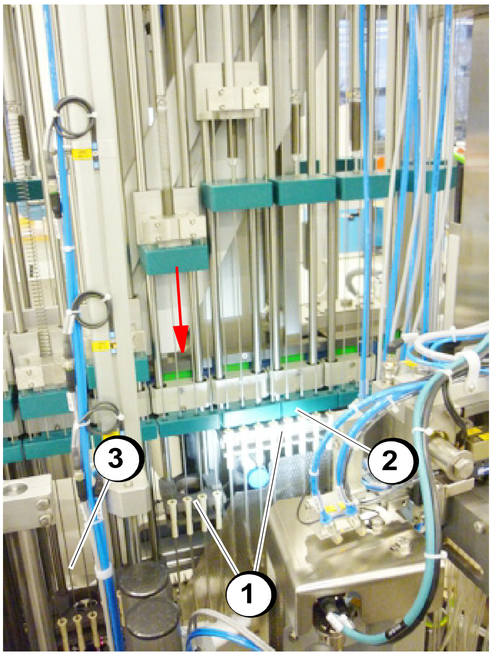
4478002B002

- Upper cam track (1) for format 40 cm
- Lower cam track (2) for format 20 cm
- Pneumatically actuated switch lever (3)

26.5.15 - am/hoe - D1_05_St_2001_en.fm

Station 2003 - Insert Catheter in Tips

A special guide ensures that the catheters are cleanly fed into the tips.



4478002B003

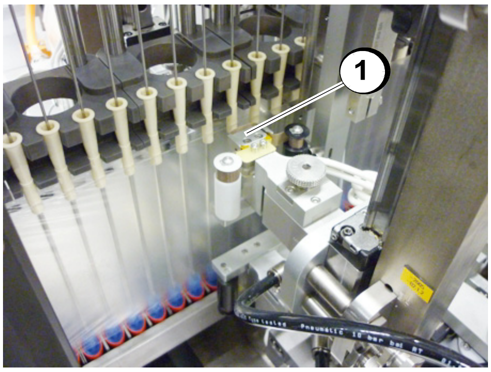
- Catheters in gripper jaws (1)
- Guide-tube holder (2)
- Mandrel (3)

The mandrels lower into the catheters and stabilize the catheter tube while being guided into the cap/tip assembly group.
The guides run another two cycles then move up again.

Station 2005 - Tack Sealing Funnel & Tip

The PU-films are connected to the funnel and insertion tip by adhesive sealing dots (impulse sealing technology).

The sealing station is mounted on the base plate in a self-supporting manner.



4478002B004

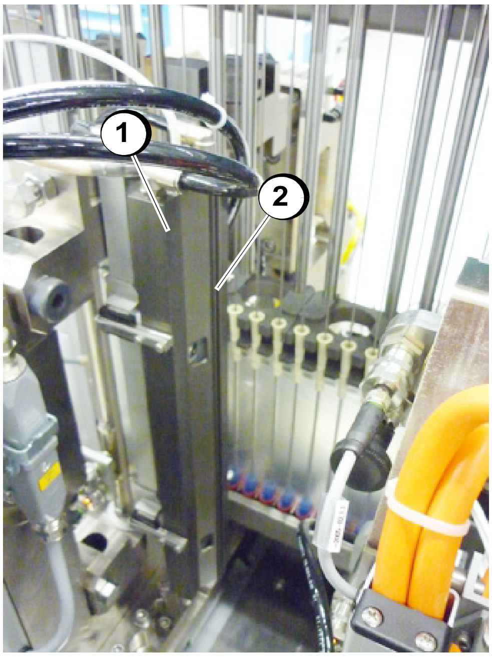
- Sealing head (1) of the cycled sealing station

Station 2011 - Long Seal

The Long Seal station seals the transverse seams over the entire width of the PU foil. The Station is assembled with a long (Format 40 cm) and a short (Format 20 cm) heated sealing tool. Appropriate to the format the station swivels 90°.

The sealing tool has a cutting edge. The sealing tool presses the foil against a rubber anvil and the cutting edge cuts the film in the middle of the sealed surfacet.

The station is mounted on the base plate in a self-supporting manner.



4478002B005

- Long sealing tool (1) swiveled 90°
- Cutting edge (2)

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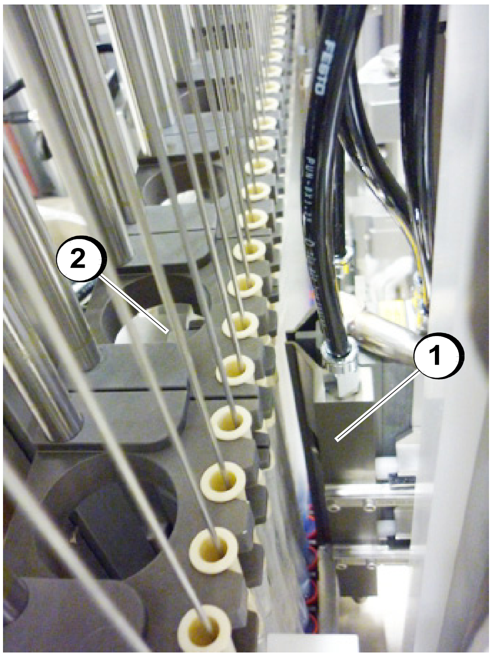


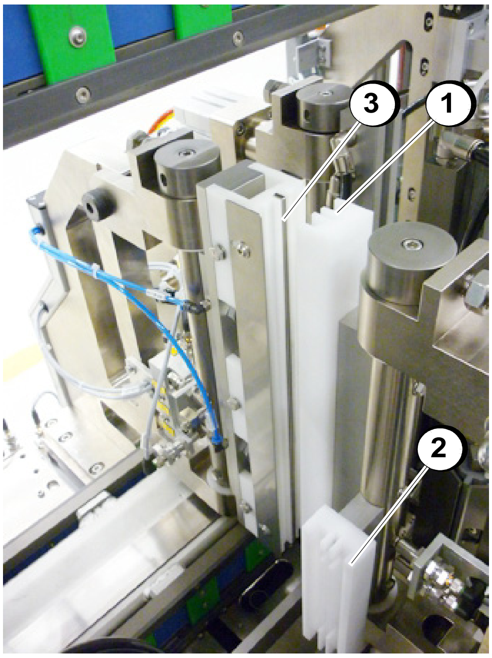
4478002B006

- Short sealing tool (1) in work position
- Rubber anvil (2)

Station 2012 - Foil Separation

The foil is clamped and cut through by a pneumatically sword.

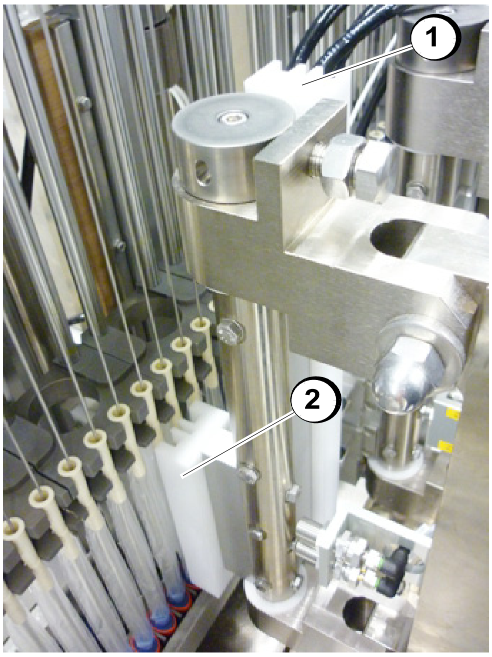
The Station is assembled with a long (Format 40 cm) and a short (Format 20 cm) clamping jaw . Appropriate to the format the station swivels 90°.



4478002B007

- Long clamping jaw (1) in working position
- Short clamping jaw (2) swiveled 90°
- Pneumatically sword (3)

26.5.15 - am/hoe - D1_05_St_2012_en.fm



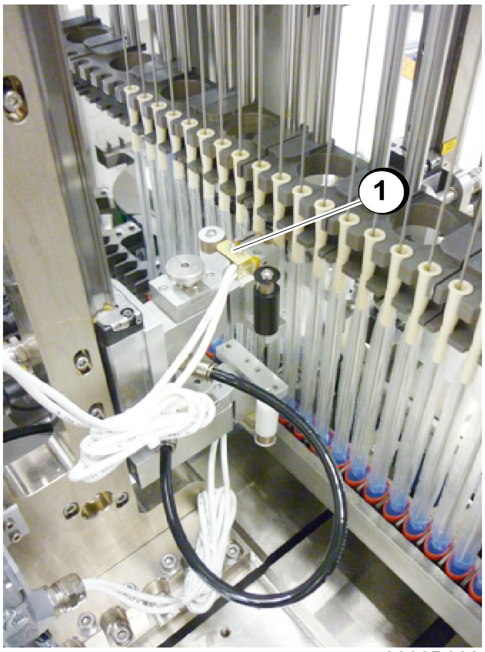
4478002B008

- Short clamping jaw (2) in working position
- Long clamping jaw (1) swiveled 90°

Station 2013 - Funnel & Tip Sealing

The PU-films are connected to the funnel and insertion tip by adhesive sealing dots (impulse sealing technology).

The sealing station is mounted on the base plate in a self-supporting manner.



4478002B009

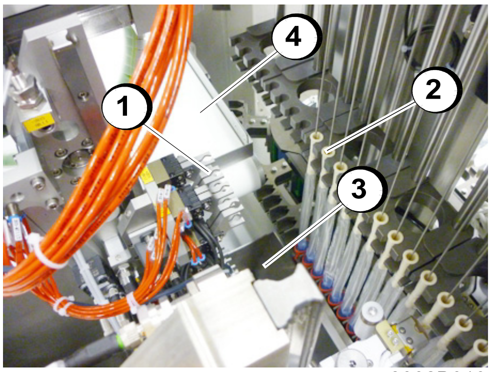
- Sealing head (1) of the cycled sealing station

26.5.15 - am/hoe - D1_05_St_2013_en.fm

Station 2015 - Pick & Place Catheters

Four catheters are picked up at the same time with special parallel grippers. The gripper pulls the catheters from the carrier and swivels them 90° to the horizontal position.

Bad parts is allowed to fall into the bad parts bin by opening the affected grippers.
The good parts are then swiveled over the discharge belt with the gripper and layed down.



4478002B010

- Pick & Place parallel gripper (1)
- Catheter in guide (2)
- Bad parts bin station 2017 (3)
- conveyor belt for good parts station 2019 (4)

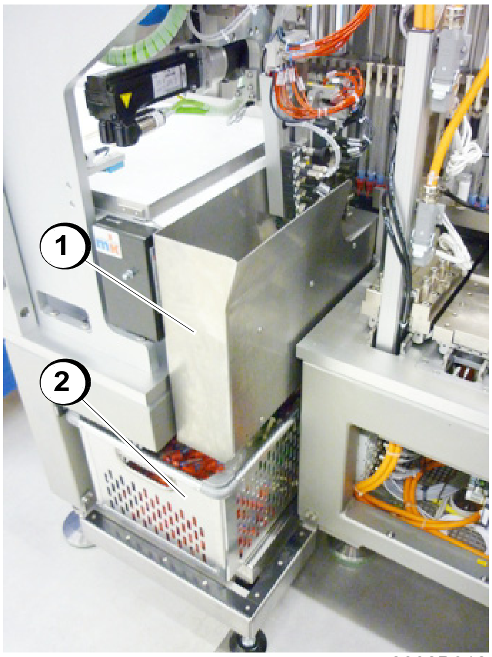
Station 2017 - Reject Station

Catheters that were declared as bad parts are dropped into the bad part bin here by the Pick & Place grippers of the previous station. The bad part inspection is performed with a scale.

The bad parts bin can be emptied during operation or while at a standstill. The machine runs with a full or missing bin until another bad part is detected.

4478002B011

- Scale (1)



4478002B012

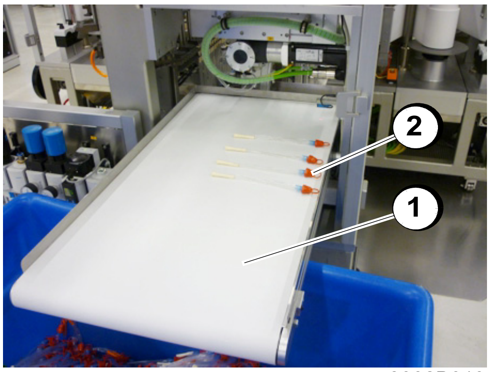
- Bad part chute (1)
- Bad parts bin (2)

Station 2019 - Discharge Conveyor

The good parts are placed on a conveyor belt.

The good parts are manually removed from the belt

.



4478002B013

- Conveyor belt (1)

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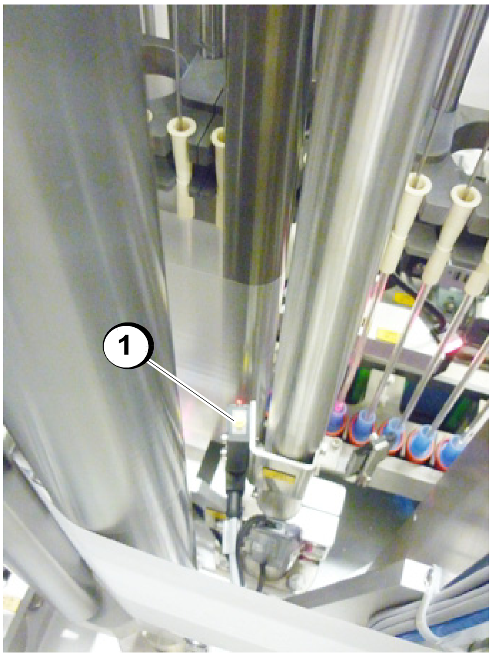
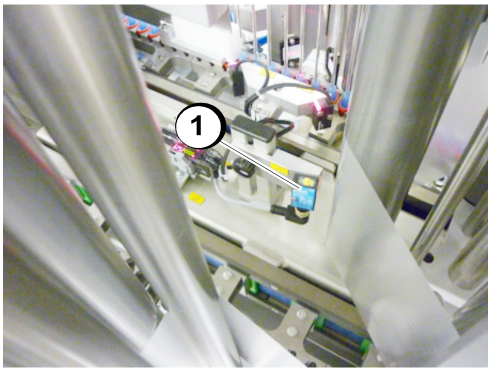
Station 2021 - Splice Detection Front Film
Station 2023 - Splice Detection Rear Film

Spice detection checks the incoming film for splices on the bottom side of the film.



4478002B014

- Sensor (1)



4478002B015

- Sensor (1)

26.5.15 - am/hoe - D1_05_St_2021_2023_en.fm

Station 2027 - Punched Catheter Feeding

The catheters are manually lifted in two linear advancer rails and placed on the funnel.

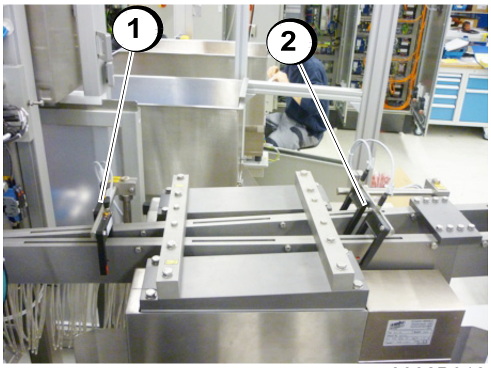
The advancer rails serve as a buffer. The total (100%) buffer capacity of 240 catheters without 2 x 22 catheters between the minimum sensor monitoring and the separator.

The catheters can be loaded in 3 minutes.

By a push button switch the feeding can be started manually.

4478002B016

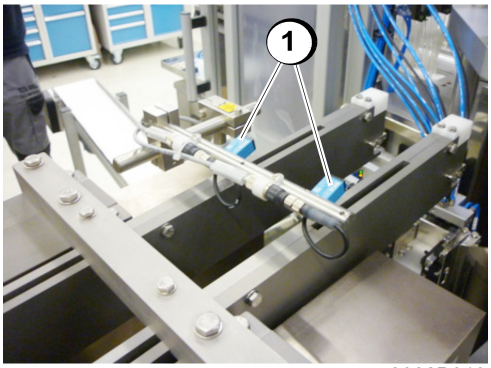
- Catheter feeding (1)



4478002B018

- Sensors „Minimal jam“ (1)
- Sensors „Prewarning minimal jam“ (2)

26.5.15 - am/hoe - D1_05_St_2027_en.fm



4478002B019

- Sensors „Maximal jam“ (1)

4478002B020

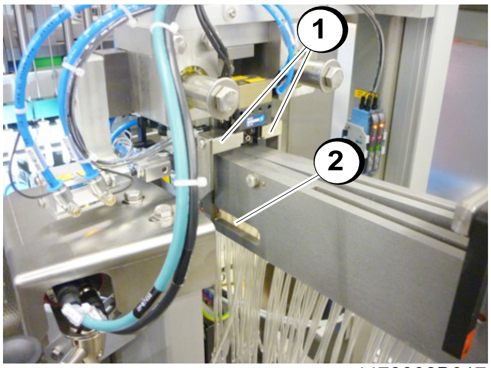
- Push button switch „Start feeding“ (1)

Station 2029 - Punched Catheter Transfer

The catheters are continuously transported into the system in two lanes. The system can, however, only process the catheters cyclically. A lock slide blocks the catheter infeed until a gripper is ready to accept the catheter. The gripper swivels approx. 180°.

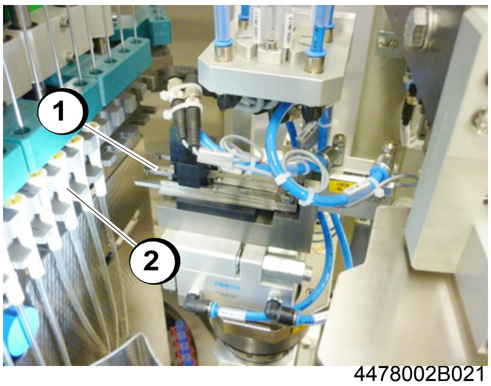
The first lock slide separates the catheters. The lock slide moves to the side and the gripper grabs two catheters. The gripper closes and the first lock slide moves to its initial position. It thereby blocks the path of the subsequent catheters.
The first lock slide retract and allow a catheter to pass through, then again block the infeed.

If the catheters are not fed in exactly, a jam may arise. To better rectify the jam, the station 2029 can be swiveled. To this a locking pin must be released.



4478002B017

- Lock slide for separation (1)
- Catheter (2)



4478002B021

- Gripper (1)
- Catheter (2)

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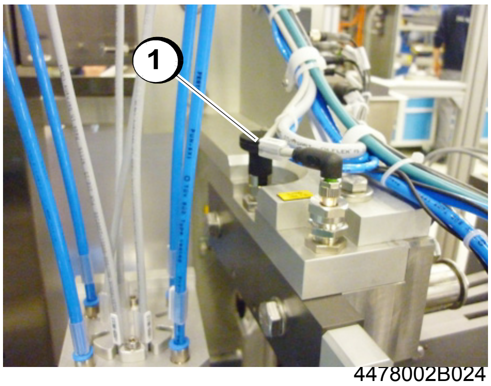


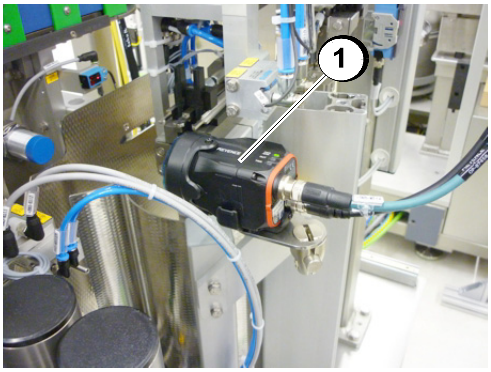
4478002B024

- Locking pin (1)

Station 2031 - CH Verification

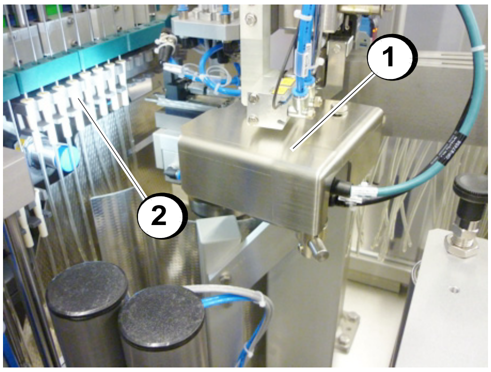
Following the separator in Station 2029, the colour camera checks the colour of the funnels.

Fals catheters are marked as bad part in the shift register and in station 2015 ejectet in the bad part bin station 2017.



4478002B022

- Colour camera (1) without cover



4478002B023

- Colour camera with cover (1)
- Catheters (2)

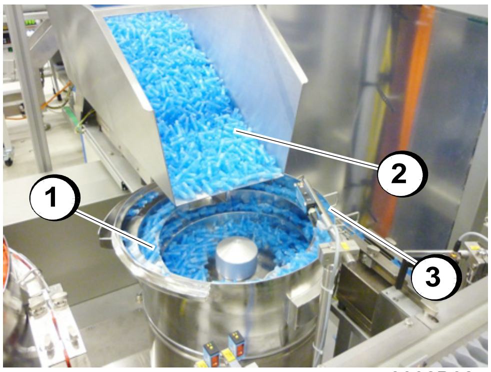
26.5.15 - am/hoe - D1_05_St_2031_en.fm

Station 2033 - Tip Feeding

The tip feeding consists of a container with a vibratory conveyor and a linear conveyor as well as discharge guides. The station transports the tips from the spiral conveyor to the downstream station.

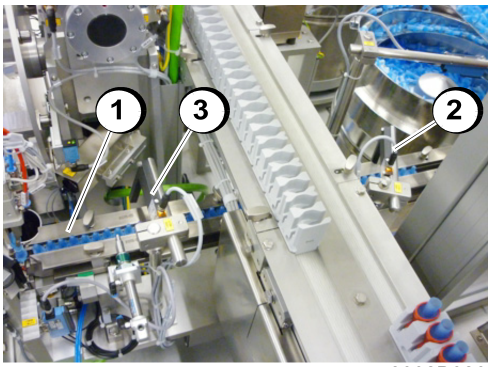
A sensor monitors for the presence of the tips in the spiral conveyor. Sensors monitor the infeed of the tips for maximum and minimum jam.

Various tips can be fed in without format adjustments.



4478002B025

- Spiral conveyor (1)
- Infeed chute (2)
- Sensor (3)



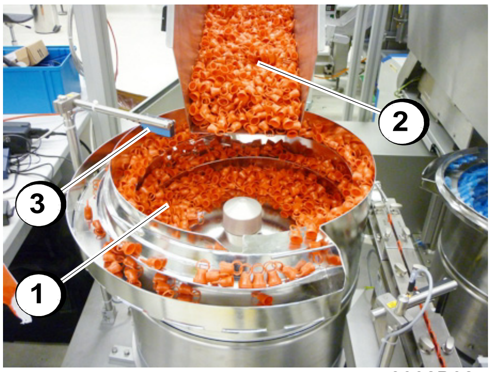
4478002B026

- Discharge guides (1) for tips
- Sensor (2) for maximum jam
- Sensor (3) for minimum jam

Station 2035 - Cap Feeding

The cap feeder consists of a bunker with a vibratory conveyor and a linear conveyor as well as a discharge guides. The station transports the caps from the spiral conveyor to the downstream station.

A sensor checks for the presence of the caps in the spiral conveyor. Sensors monitor the infeed of the caps for maximum and minimum jam.



4478002B027

- Spiral conveyor (1)
- Infeed chute (2)
- Sensor (3)

4478002B028

- Discharge guides (1) for caps
- Sensor (2) for maximum jam
- Sensor (3) for minimum jam

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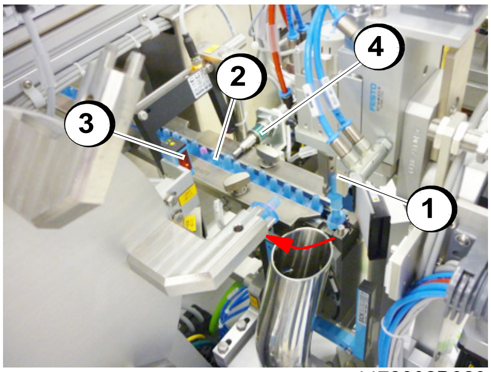
Station 2037 - Insert Tip

The separated tips are picked up by a gripper, rotated 90° after the camera inspection (station 2039)and plugged on the mounting wheel of the station 2043. Bad parts are discharged in the ejection chute.

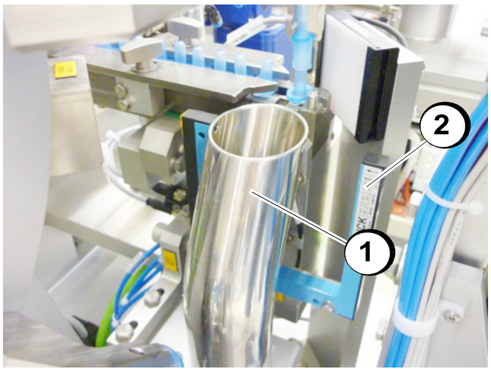
A stopper is reducing the back pressure in the infeed rail in front of the tip selection.

A sensor ensures that a tip was grabbed.



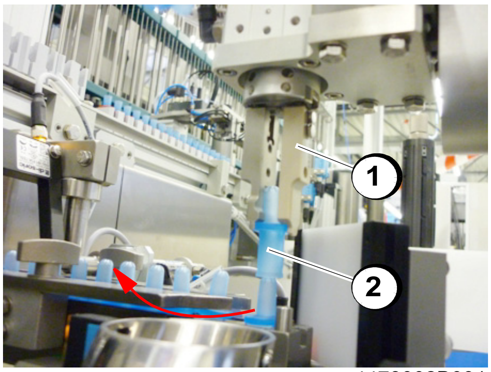
4478002B029

- Gripper (1)
- Tips in infeed (2)
- Presence sensor (3) „Tip in mounting wheel“
- Stopper (4)



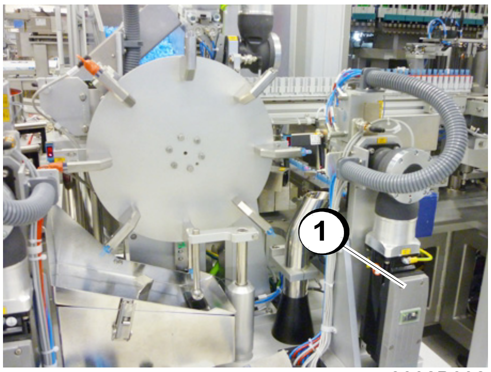
4478002B030

- Ejection chute (1)
- Sensor „Bad part ejected“ (2)



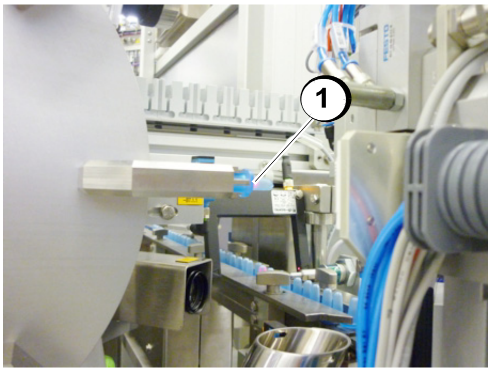
4478002B031

- Gripper (1)
- Tip (2)



4478002B032

- Servo (1) for turning 90°



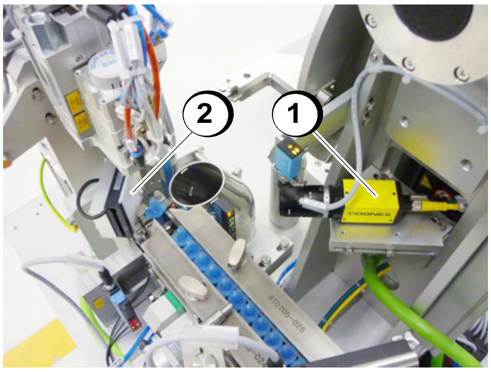
4478002B033

- Mounting wheel of the station 2043 (1)

26.5.15 - am/hoe - D1_05_St_2037_en.fm

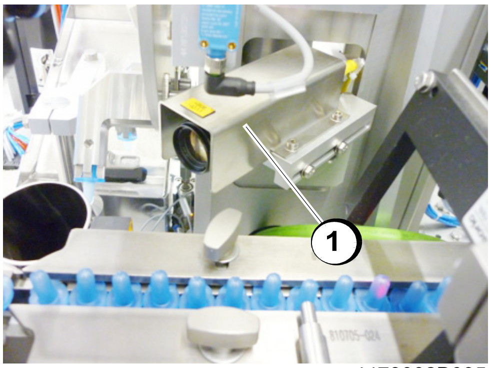
Station 2039 - Vision System Tip

The tips are checked by a camera for size and completeness.



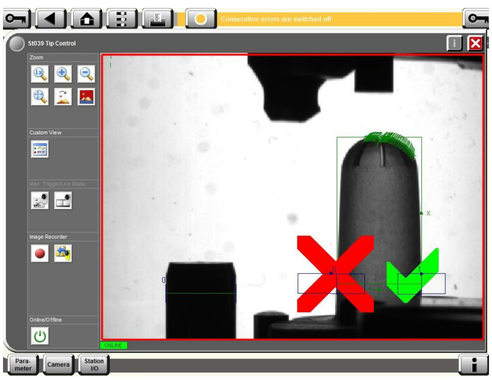
4478002B034

- Camera (1) without cover
- Background lighting (2)



4478002B035

- Camera with cover (1)



vision_system

Camera image
Checked are:

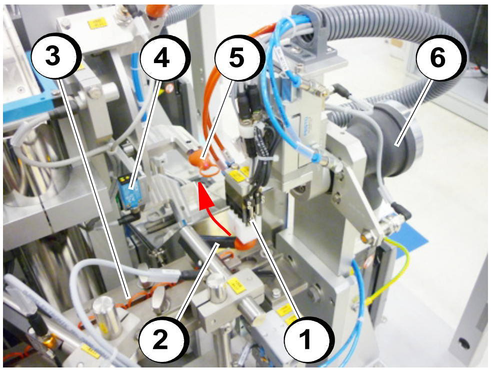
- Tip diameter
- Tip incompleteness check
- Pin-diameter reference

Observe!

For information on the operation, function and adjustment, please refer to the manufacturer's operating manual.

Station 2041 - Insert Cap

A gripper fetches the caps from the separator, turns 90° and place them on the tips in the turntable of station 2043.



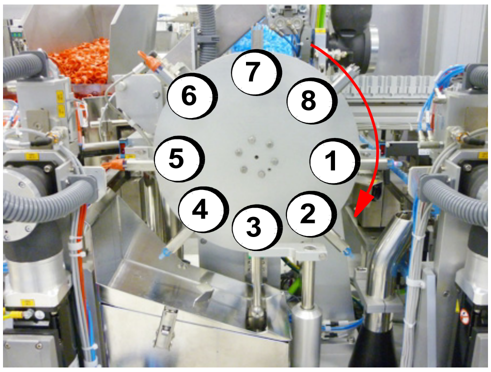
4478001B013

The cap is lifted from the infeed belt by a gripper and inserted onto the tip.

- Gripper (1)
- Sensor (fork light barrier) checks presence of cap in pick position (2)
- Infeed chute (3)
- Sensor checks the infeeding of the cap onto the tip (4)
- Cap (5)
- Servo (6)

Station 2043 - Cap/Tip Assembly (incl. Lubragel Container)

On a vertical turntable are 4stations mounted. Additional are 4 empty stations on the turntable.

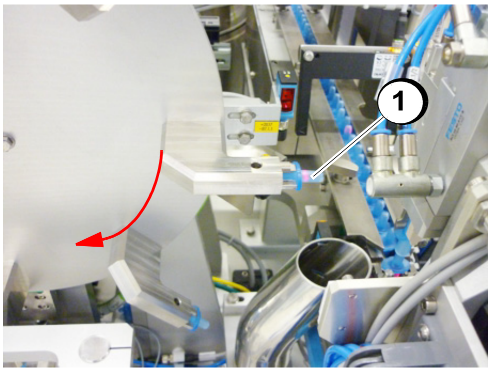


4478002B037

Turntable with 8 positions:

- Transfer station (1)
- Empty station (2)
- Lubricating station (3)
- Empty station (4)
- Mounting cap/tip station (5)
- Empty station (6)
- Remove cap/tip (7)
- Empty station (8)

Transfer tip from Station 2037



4478002B038

The tip is inserted at the 3 o'clock position with the opening forward faeing.

- Tip (1)

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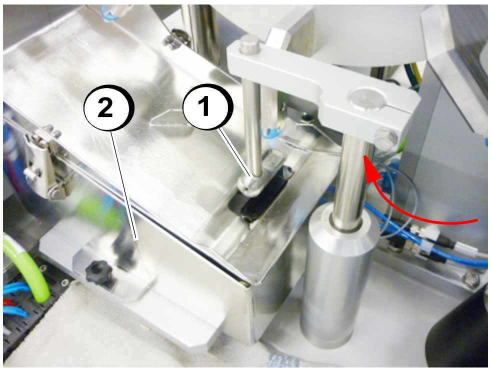
Lubricating station

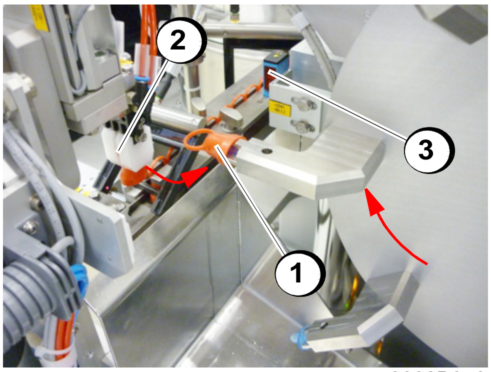


4478002B039

At the 6 o'clock position, the tip is dipped into a lubricating fluid (Surplus Lubragel) with a lubricating head.

- Lubricating head
- Dip tank

Assembly station

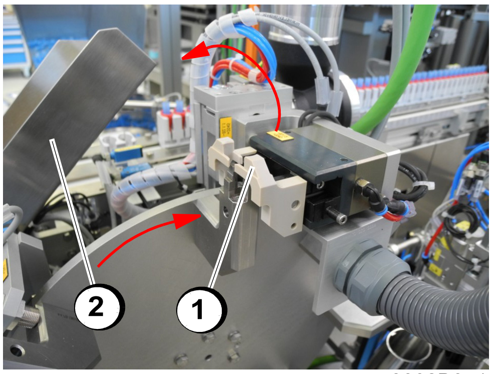


4478002B040

At the 9 o'clock position, the cap is I inserted onto the tip.

- Cap onto Tip (1)
- Gripper - cap infeed (2)
- Presence sensor (3)

Unloading station



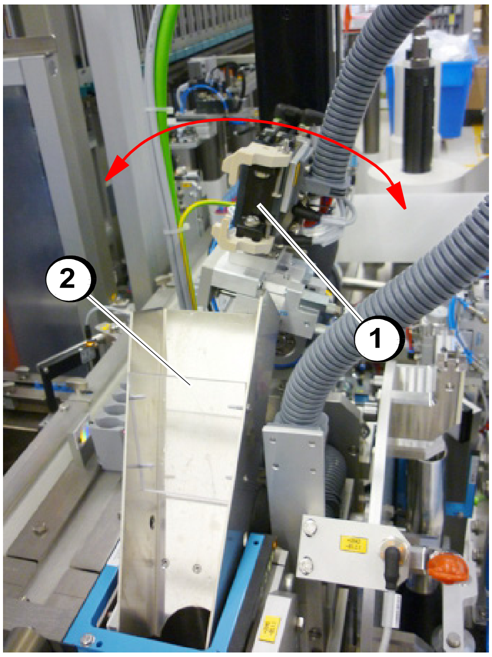
4478002B041

At the 12 o'clock position, the assembled caps/tips are removed from the turntable and placed in the pucks of the infeed belt.

- Gripper (1)
- Bad part chute (2)

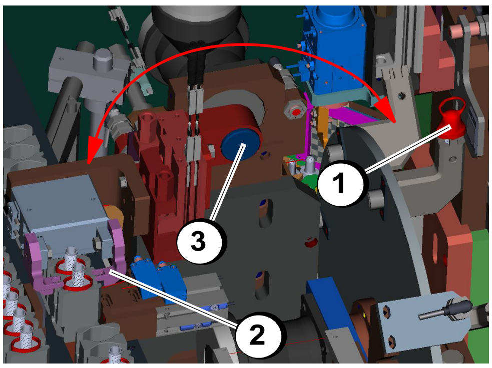
Station 2045 - Cap/Tip Transfer

The assembled caps/tips are removed from the turntable with a gripper. The gripper rotates vertically 180° and places the assembled caps/tips on the pucks of the infeed belt.



4478002B042

- Gripper (1)
- Bad parts chute (2)



2045_neu

- Assembled caps/tips on turntable (1)
- Gripper (2)
- Pivot point (3)

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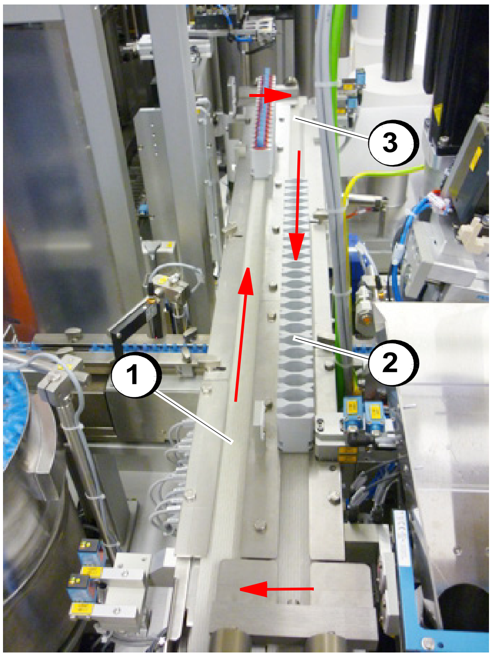
Station 2047 - Load pucks/buffer
Station 2048 - Unload pucks

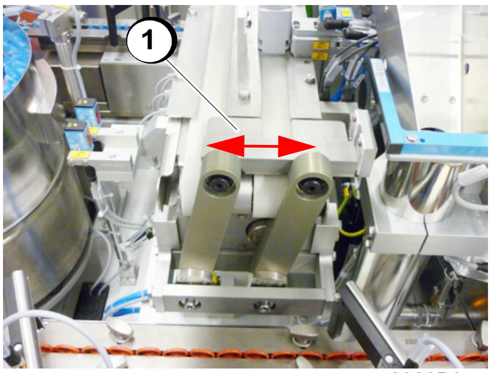
The infeed belt with pucks has the function of a buffer with a capacity of approx. 2 production minutes. The empty pucks are transported back on the reverse transport belt. The cross transport of the pucks is performed by pneumatical pushers.

Pick and place of the Cap/Tip is controlled by sensors.



4478002B043

- Infeed belt (1)
- Pucks (2)
- Reverse transport belt (3)



4478002B044

- Pneumatical pusher (1)
-

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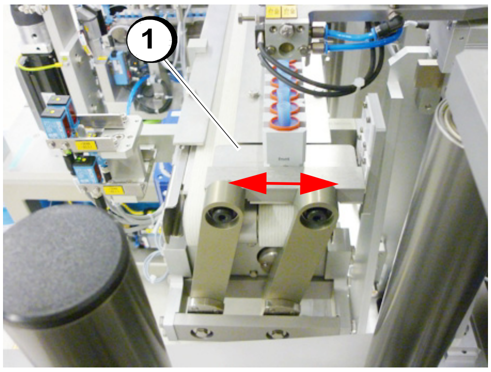
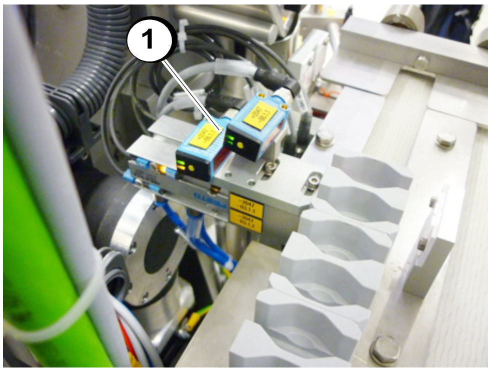


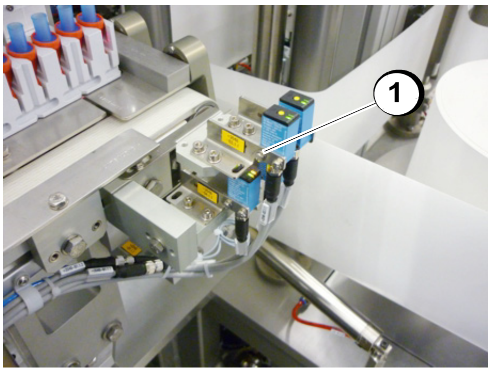
4478002B045

- Pneumatical pusher (1)



4478002B046

- Sensor Place (1)



4478002B047

- Sensors Pick (1)

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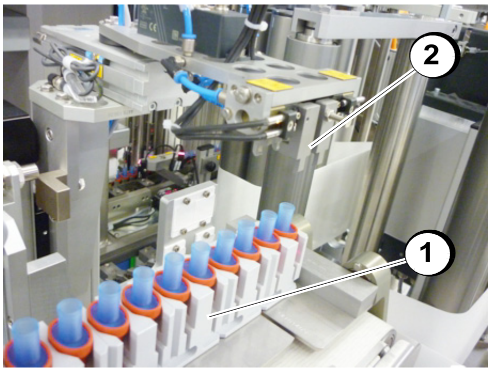
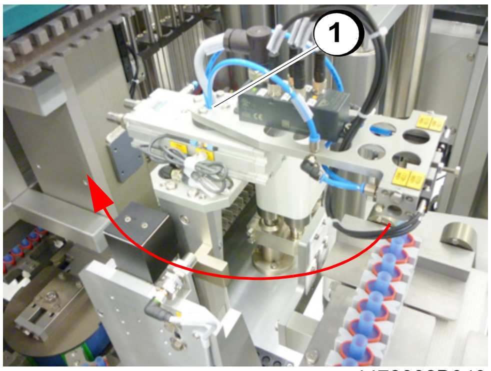
Station 2049 - Insert Cap/Tip

Two assembled caps/tips of a puck are picked up by a gripper. The gripper rotates horizontally 180° and place the two assembled caps/tips in the cap/tip carrier of the oval transport system. A sensor monitors the successful transfer into the cap/tip carrier.



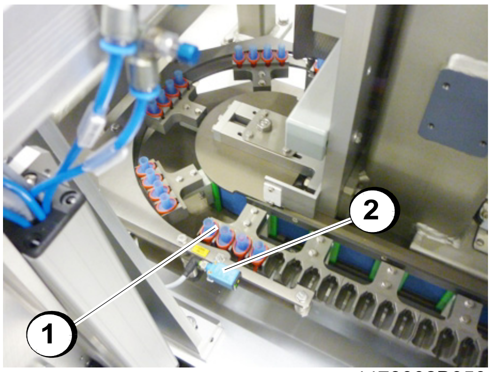
4478002B048

- Pucks (1)
- Gripper (2)



4478002B049

- Pivot point (1)



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- Cap/tip carrier in the oval transport system (1)
- Sensor (2)

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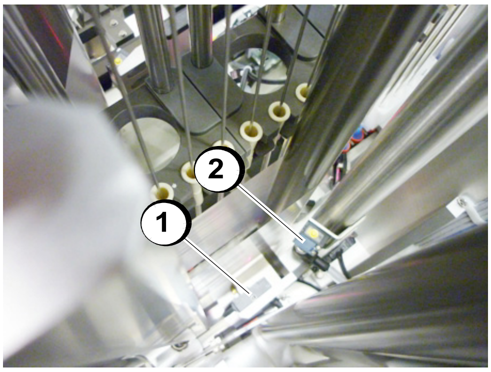
Station 2051 - Position Verification Front Film
Station 2053 - Position Verification Rear Film

The respective position of the film is checked forward the sealing station with a laser sensor.



4478002B051

Station 2051/2053 Film web:

- Sensor for the detection of the film (1)
- Sensor Splice detection (2)

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Station 2101 - Edge Guiding Front Film
Station 2103 - Edge Guiding Rear Film

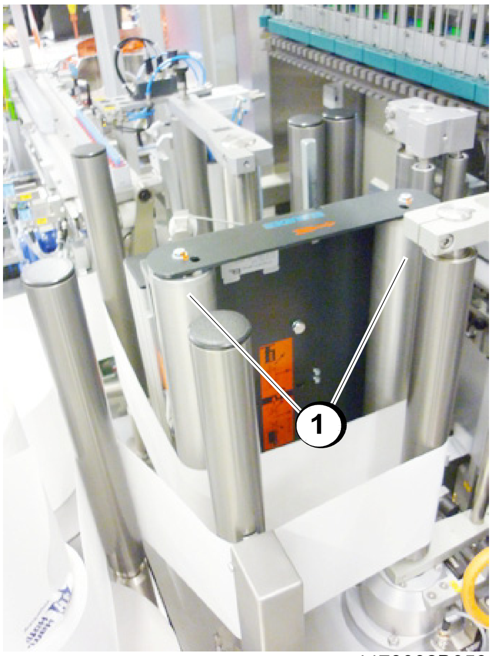
A web edge control corrects wandering of the film web.

- The web path is corrected by adjusting the axes of a pair of reels (perpendicular to the direction of film transport).
- Sensors detect the web edge and output signals for their correction.

4478002B052

Web edge control of station 2101
Film web:

- Adjustable roller pair (1)



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Web edge control of station 2103
Film web:

- Adjustable roller pair (1)

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Station 2150 - Unwind Front Film
Station 2151 - Unwind Rear film

- The station consists of:
- Servo-driven pneumatic expansion shaft
- Dancer
- Pneumatic gluing table
- Ionization

The film is constantly unwound from the expansion shaft.

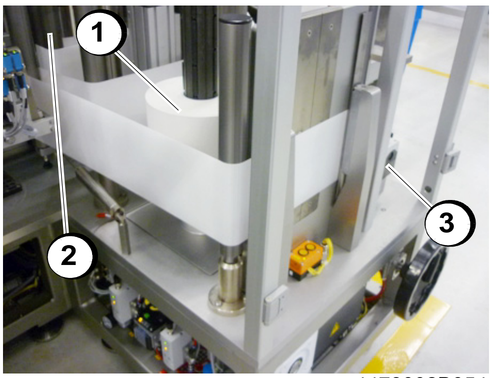
The required web tension and the control of the unwinding process is performed via a dancer control.

The station contains a pneumatic gluing table.

The reel spindle can be manually swiveled to the horizontal position for changing the reel.

Unwinding with servo

- The pneumatic expansion shaft is mounted in bearing cap.
- Unwinding is directly driven via a servo motor with gearing.
- Tensioning of the expansion shaft is performed by a switch.
- Unwinding speed is determined by the dancer position.
- Prewarning and detection of "reel end" by calculating the roll diameter.



4478002B054

Example unwinding at station 2150:

- Unwinding (1)
- Dancer (2)

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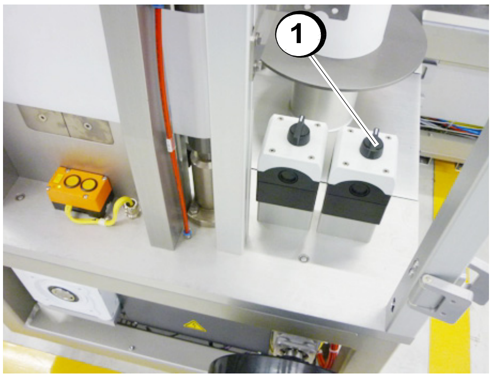
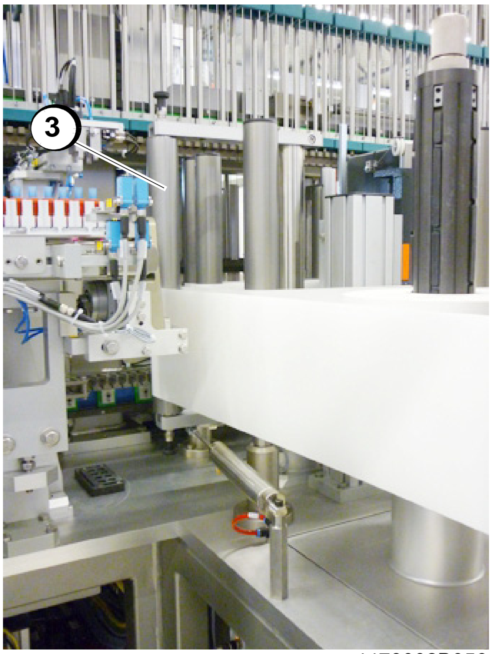


4478002B055

- Switch for tensioning or releasing the reel spindle for winding

Dancer The dancer arm with rocker mounting controls unwinding (located in front of it) depending on the position.

- The film tension is pneumatically generated by a low-friction cylinder and can be set at the HMI via a pressure control value.
- Position detection of the dancer arm is performed without contact by means of an analog sensor.
- Detection of "Winding speed too fast / too slow" by comparing dancer position with winder drive.
- Detection of film tear by means of the dancer position.



4478002B056

- Dancer (1)

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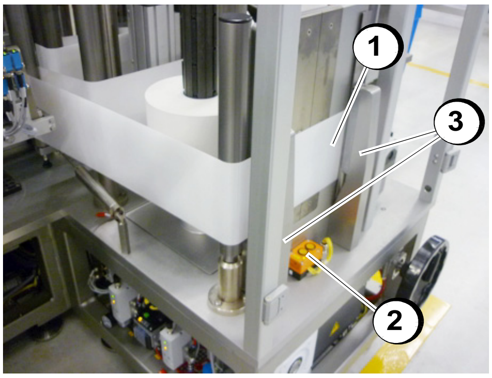
Gluing table

On the gluing table the foil is glued (e.g. by foil tear or new foil roll).

The pneumatic clamping bar fixes the film to the new reel during splicing. The clamping bar can be manually opened and closed via the button



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- Gluing table (1)
- Button (2) for opening and closing the clamping bar (3)
- Guide rails (4)

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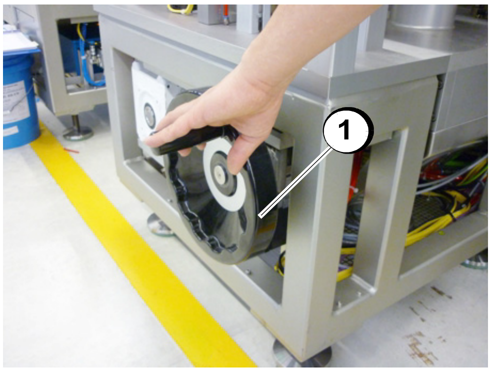
Expansion shaft swiveled

Expansion shaft swiveled by the hand wheel in position for changing the roll



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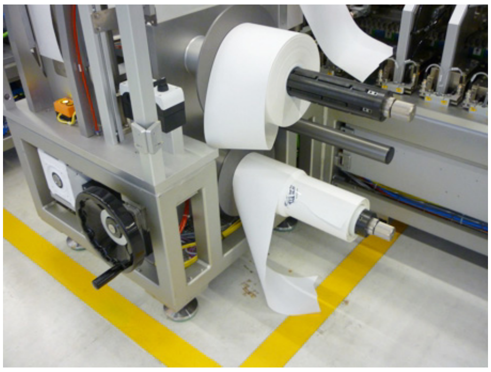
- Hand wheel (1)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→ Expansion shaft swiveled 90° (1)

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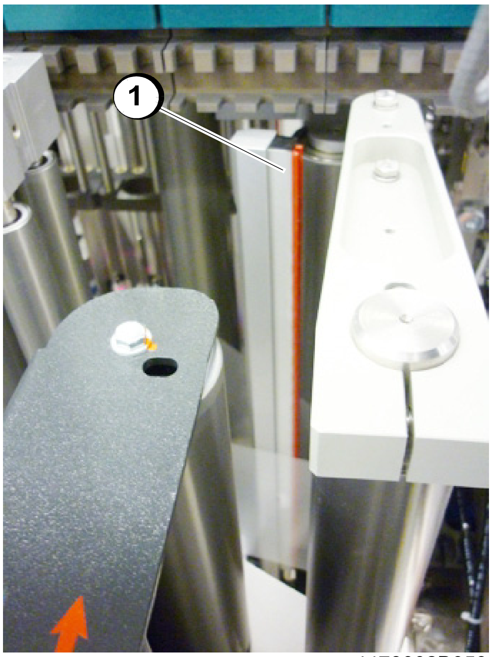
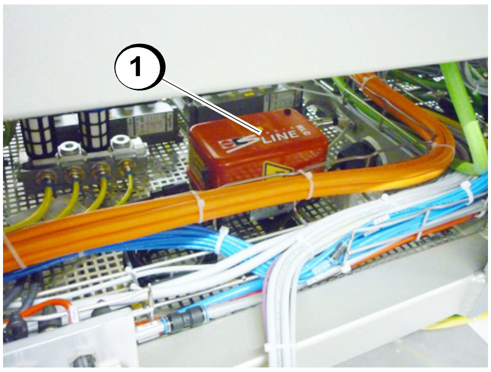
Ionisation Ionization eliminates electrostatic charging on the foil.



4478002B059

- Ionization bars (1)



4478002B060

- Power supply unit (1)

Observe!

For information on the operation, function and adjustment, please refer to the manufacturer's operating manual.

26.5.15 - am/hoe - D1_05_St_2150_2151_en.fm

Station 2170 - Rewind Front Film Carrier
Station 2171 - Rewind Rear Film Carrier

The station consists of:

- Servo-driven pneumatic expansion shaft
- Dancer
- Pneumatic claming jaw

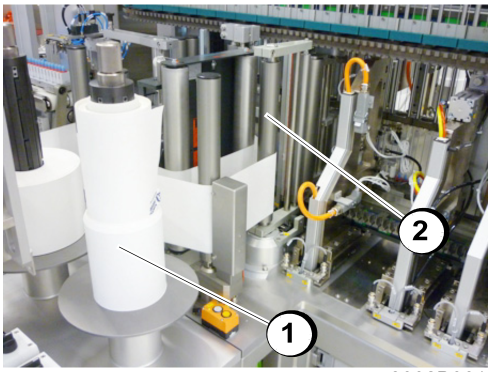
The film is constantly rewound on the expansion shaft.

The required web tension and the control of the rewinding process is performed via a dancer control.

The reel spindle can be manually swiveled to the horizontal position for changing the reel.

Rewinding with servo

- The pneumatic expansion shaft is mounted on in bearing cap.
- Rewinding is directly driven via a servo motor with gearing.
- Tensioning is performed by a side valve connection.
- Rewinding speed is determined by the dancer position.
- Prewarning and detection of "reel end" by calculating the roll diameter.



4478002B061

Example rewinding at station 2170:

- Rewinding (1)
- Dancer (2)

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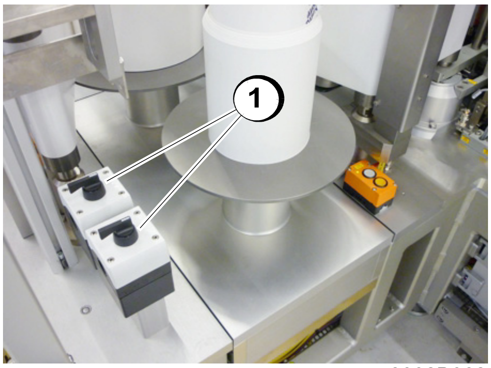


4478002B062

- Button (1) for tensioning or releasing the expansion shafts

Dancer

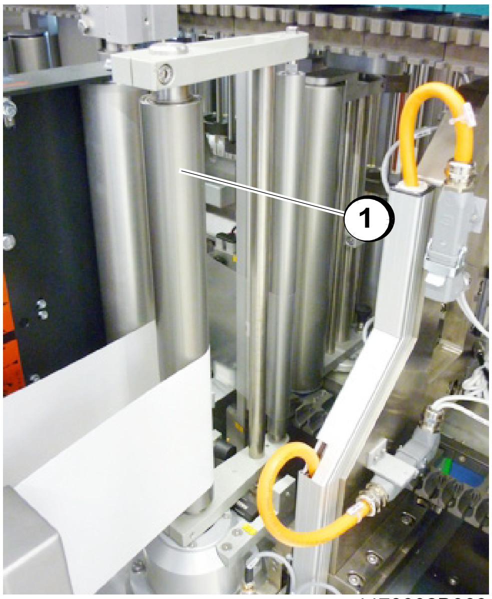
The dancer arm with rocker mounting controls rewinding (located in front of it) depending on the position.

The film tension is pneumatically generated by a low-friction cylinder and can be manually set via a pressure control value.

Position detection of the dancer arm is performed without contact by means of an analog sensor.

Detection of "Winding speed too fast / too slow" by comparing dancer position with winder drive.

Detection of film tear by means of the dancer position.



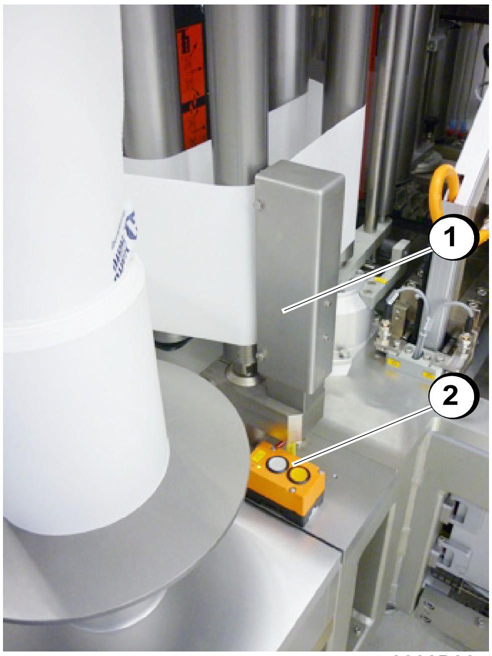
4478002B063

- Dancer (1)

Claming jaw

The clamping jaw is fixing the foil (e.g. by foil tear or new foil roll).

The clamping jaw can be manually opened and closed via the button



4478002B064

- Clamping jaw (1)
- Button (2) for opening and closing the clamping jaw

26.5.15 - am/hoe - D1_05_St_2170_2171_en.fm